Levels 3-4: Applying/Analyzing

4. Which of the following statements is true?
(A) An antibody has one antigen-binding site.
(B) A lymphocyte has receptors for a single antigen.
(C) Every antigen has a single epitope.
(D) A liver or muscle cell makes two classes of MHC molecule.

5. Which of the following should be the same in identical twins?
(A) the set of antibodies produced
(B) the set of MHC molecules produced
(C) the set of T cell antigen receptors produced
(D) the set of immune cells eliminated as self-reactive

Levels 5-6: Evaluating/Creating

6. Vaccination increases the number of
(A) different receptors that recognize a pathogen.
(B) lymphocytes with receptors that can bind to the pathogen.
(C) epitopes that the immune system can recognize.
(D) MHC molecules that can present an antigen.

7. Which of the following is least likely to help a virus avoid triggering an adaptive immune response?
(A) having frequent mutations in genes for surface proteins
(B) infecting cells that produce very few MHC molecules
(C) producing proteins very similar to those of other viruses
(D) infecting and killing helper T cells

8. **DRAW IT** Consider a pencil-shaped protein with two epitopes, Y (the "eraser" end) and Z (the "point" end). They are recognized by antibodies A1 and A2, respectively. Draw and label a picture showing the antibodies linking proteins into a complex that could trigger endocytosis by a macrophage.

9. **MAKE CONNECTIONS** Contrast clonal selection with Lamarck's idea for the inheritance of acquired characteristics (see Concept 22.1).

10. **EVOLUTION CONNECTION** Describe one invertebrate mechanism of defense against pathogens and discuss how it is an evolutionary adaptation retained in vertebrates.

11. **SCIENTIFIC INQUIRY** A major cause of septic shock is the presence in blood of lipopolysaccharide (LPS) from bacteria. Suppose you have available purified LPS and several strains of mice, each with a mutation that inactivates a particular TLR gene. Explain how you might use these mice to test the feasibility of treating septic shock with a drug that blocks TLR signaling.

12. **WRITE ABOUT A THEME: INFORMATION** Among all nucleated body cells, only B and T cells lose DNA during their development and maturation. In a short essay (100–150 words), discuss the relationship between this loss and DNA as heritable biological information, focusing on similarities between cellular and organismal generations.

13. **SYNTHESIZE YOUR KNOWLEDGE**

This child is receiving an oral vaccine against polio, a disease caused by a virus that infects neurons. Given that the body cannot replace most neurons, why must a polio vaccine stimulate both a cell-mediated and a humoral response?

For selected answers, see Appendix A.

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